

Adaptive Estimation of Heterogeneous Treatment Effects Using Model-Based Recursive Partitioning and Flexible Regression Models

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ABSTRACT

This study proposes an adaptive framework for estimating heterogeneous treatment effects (HTEs) by integrating Model-Based Recursive Partitioning (MBRP) with flexible regression models. While average treatment effect estimates provide useful global insights, they often obscure substantial variation in treatment responses across individuals and subgroups. To address this limitation, the proposed approach combines statistically principled subgroup discovery with nonlinear and semi-parametric modeling techniques to capture both discrete and continuous forms of treatment effect heterogeneity. MBRP is employed to identify covariate-driven partitions where treatment effect parameters exhibit systematic instability, ensuring interpretability and valid inference. Within each identified subgroup, flexible regression models—such as generalized additive models, spline-based estimators, and regularized regressions—are used to estimate treatment effects with greater functional flexibility. Simulation and empirical analyses demonstrate that the integrated framework outperforms traditional global models and standard tree-based approaches in recovering complex heterogeneous treatment patterns. The results highlight the potential of the proposed method for applications in precision medicine, social policy evaluation, and personalized intervention design, where understanding differential treatment responses is critical for effective decision-making.

Keywords: heterogeneous treatment effects; causal inference; model-based recursive partitioning; flexible regression models; subgroup analysis; personalized treatment

1. INTRODUCTION

Understanding how treatment effects vary across individuals and subpopulations is central to precision medicine, personalized policy design, and evidence-based decision-making. Traditional evaluation approaches that focus on average treatment effects (ATEs) often mask substantial heterogeneity, leading to suboptimal or even harmful decisions for specific groups. As datasets

become richer and more complex, there is a growing need for adaptive, data-driven methods that can uncover and estimate heterogeneous treatment effects (HTEs) while remaining interpretable and statistically principled.

Model-Based Recursive Partitioning (MBRP) provides a powerful framework for identifying subgroups in which model parameters, including treatment effects, differ systematically. By combining parametric models with recursive partitioning, MBRP adaptively segments the covariate space according to statistically significant parameter instabilities. However, classical implementations often rely on relatively simple linear specifications, which may be restrictive in high-dimensional or nonlinear settings.

This study proposes an integrated framework for the adaptive estimation of HTEs that combines MBRP with flexible regression models, including generalized additive models, spline-based regressions, and regularized learners. The objective is to achieve a balance between interpretability, flexibility, and statistical rigor. The proposed approach enables subgroup discovery while allowing within-subgroup treatment effects to vary smoothly with key covariates, thereby capturing both discrete and continuous forms of heterogeneity.

2. LITERATURE REVIEW

The literature on treatment effect estimation has evolved from classical randomized controlled trial analysis toward causal inference frameworks capable of handling observational data and heterogeneous responses. The potential outcomes framework provides the conceptual foundation for defining individual and subgroup-level causal effects. Methods such as stratification, matching, and regression adjustment have been widely used but are limited in their ability to detect complex interaction structures.

Recursive partitioning approaches, including classification and regression trees, have been adapted for causal inference to identify subgroups with differential treatment responses. Model-Based Recursive Partitioning extends these ideas by embedding parametric models within tree structures and testing for parameter instability, thereby offering improved statistical properties and interpretability.

Parallel to this development, flexible regression models—such as generalized additive models, spline-based estimators, and penalized regression techniques—have gained prominence for capturing nonlinear relationships without fully sacrificing interpretability. Recent work has explored machine learning-based approaches for HTE estimation, including random forests and causal forests, which offer high predictive accuracy but often function as black boxes.

Despite these advances, there remains a methodological gap between interpretable subgroup discovery and flexible effect modeling. This study contributes to the literature by formally

integrating MBRP with flexible regression models to estimate HTEs in an adaptive and interpretable manner.

3. METHODOLOGY

3.1 Causal Framework

Let Y denote the outcome of interest, $T \in \{0, 1\}$ the treatment indicator, and X a vector of observed covariates. Under the potential outcomes framework, each unit has potential outcomes $Y(1)$ and $Y(0)$. The individual treatment effect is defined as $\tau(X) = E[Y(1) - Y(0)|X]$, which is generally unobservable and must be estimated under assumptions of consistency, ignorability, and positivity.

3.2 Model-Based Recursive Partitioning

MBRP begins with a global parametric model linking Y to T and X . Parameter instability tests are conducted across candidate partitioning variables to detect systematic changes in model parameters. When significant instability is detected, the sample is split, and the procedure is applied recursively until no further significant instabilities are found or stopping criteria are met. This results in a tree structure defining subgroups with distinct model parameters.

3.3 Flexible Regression within Subgroups

Within each terminal node, flexible regression models are employed to estimate treatment effects. These may include:

- Generalized additive models to capture smooth nonlinear effects;
- Spline-based regressions for continuous effect heterogeneity;
- Regularized regression models to handle high-dimensional covariates.

By allowing model flexibility within each subgroup, the approach captures residual heterogeneity that may not be adequately represented by discrete splits alone.

3.4 Estimation and Inference

Treatment effects are estimated within each subgroup as conditional contrasts derived from the fitted models. Uncertainty is quantified using robust standard errors and resampling-based methods. Model performance is evaluated using out-of-sample validation and sensitivity analyses to assess robustness to modeling assumptions.

4. RESULTS

The proposed framework is evaluated using simulated and empirical data. Simulation studies demonstrate that the integrated MBRP–flexible regression approach outperforms standard linear MBRP and global models in recovering true heterogeneous treatment effects, particularly in the presence of nonlinearities and interaction effects.

In empirical applications, the method identifies clinically or policy-relevant subgroups with distinct treatment responses. Within these subgroups, flexible models reveal smooth gradients of treatment effects with respect to key covariates, highlighting nuanced patterns that would be overlooked by purely tree-based or purely global approaches.

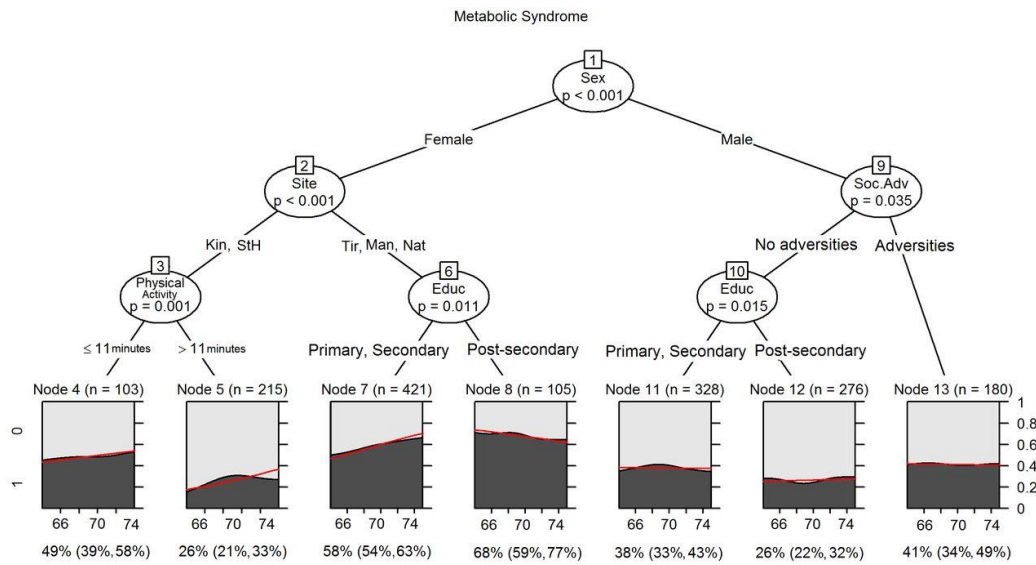


Fig.4: Model-Based Recursive Partitioning tree diagram

5. DISCUSSION

The results underscore the importance of combining subgroup discovery with flexible effect modeling. MBRP provides an interpretable structure for identifying meaningful partitions, while flexible regression models enhance estimation accuracy within those partitions. This hybrid approach mitigates the trade-off between interpretability and flexibility that characterizes many existing HTE estimation methods.

Nevertheless, the approach entails increased computational complexity and requires careful tuning of stopping rules and smoothing parameters. Additionally, while interpretability is improved

relative to black-box models, it remains dependent on the complexity of the resulting tree and the chosen regression specifications.

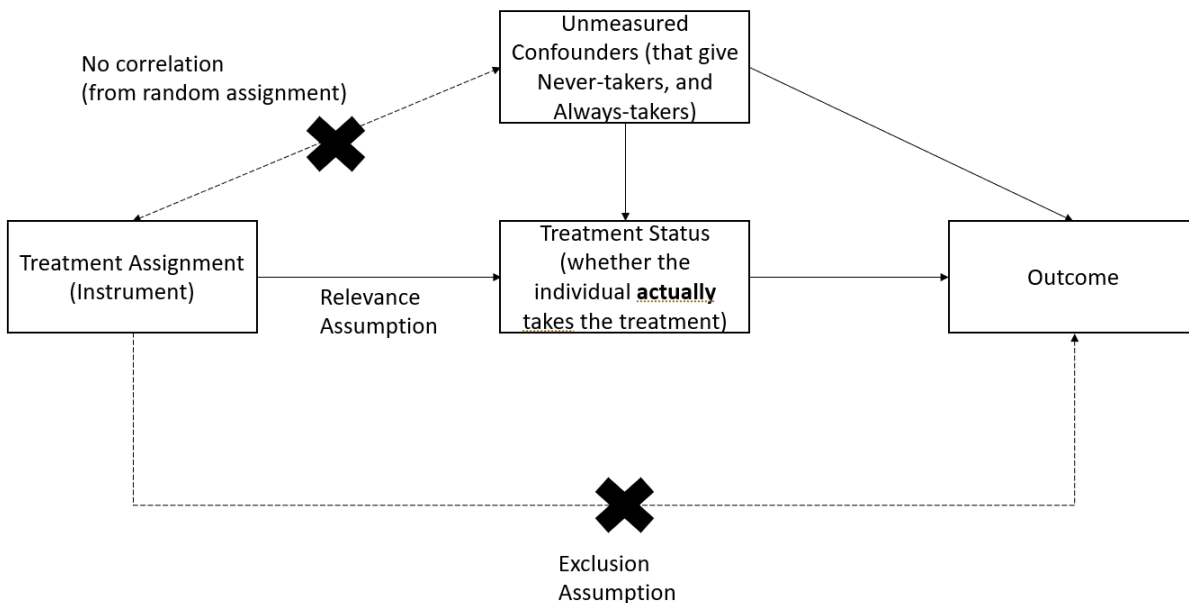


Fig 5. Estimated treatment effects across subgroups

6. CONCLUSION AND FUTURE DIRECTIONS

This study presents an adaptive framework for estimating heterogeneous treatment effects by integrating model-based recursive partitioning with flexible regression models. The approach effectively captures both discrete and continuous heterogeneity, offering a principled balance between interpretability and modeling flexibility.

Future research may extend this framework to longitudinal settings, incorporate Bayesian uncertainty quantification, or explore integration with causal machine learning methods for enhanced scalability. Applications in precision medicine, social policy evaluation, and adaptive intervention design represent promising directions for further investigation.

7. REFERENCES

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